AI-Driven Multi-Modal Adaptive Handover Control for O-RAN

thesis

2026-06-23

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	AI-Driven Multi-Modal Adaptive Handover Control Optimization for O-RAN Abdul Wadud, Fatemeh Golpayegani, Nima Afraz (University College Dublin) arXiv:2603.17158v1 · IEEE ICC 2026 Workshop (WS-01) O-RAN, handover, multi-modal mobility, Non-RT/Near-RT RIC, RL

. (7) · RSRP , PPO non-RT RIC **rApp** , near-RT RIC **xApp** E2 hierarchical mobility-aware handover .

2 Motivation

.

(1) A3 mobility-agnostic . Traditional A3-event handover TTT / Hysteresis / CIO . robust , · . fading ping-pong handover , late/failed handover RLF . · .

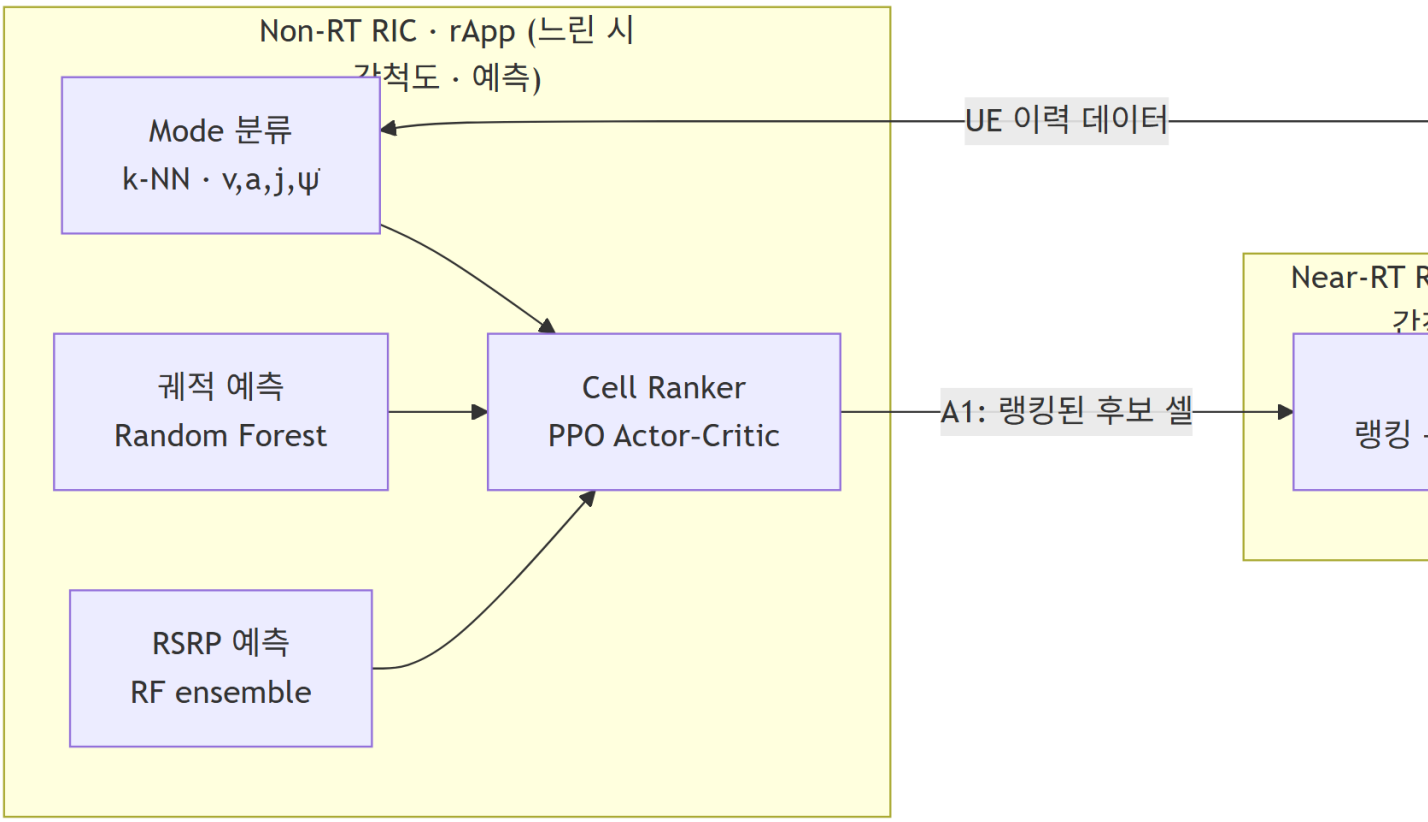
(2) ML . · · · · **multi-modal** , ML handover .

(3) near-RT . DQN/LSTM near-RT RIC .

: multi-modal + + O-RAN .

3 System Model & Architecture

(rApp, non-RT) (xApp, near-RT) , **A1** .



. · non-RT rApp , near-RT xApp .
sub-second RIC .

4

4.1 (MC)

$(v, a, j, \dot{\psi}) +$ **k-NN** . jerk $j = d^3x/dt^3($, $)$ bearing rate $\dot{\psi} = v\kappa($ $)$
feature . 7 {ped, cyclist, car, bus, train, drone, uav} one-hot $\mathbf{e}(m_t^u)$.

4.2 (TP) · RSRP (P-RSRP)

Random Forest .

- **TP**: (v, a, m, x, y) $\hat{p}_{t+1}^u$
- **P-RSRP**: feature RSRP $\hat{r}_{t+1}^u$

LSTM/DQN "near-RT "

4.3 Cell Ranker (CR) — PPO Actor–Critic

- **state**: $s_t^u = [\,\mathbf{e}(m_t^u),\ \widehat{p}_{t+1}^u,\ \widehat{r}_{t+1}^u\,]$ — + + RSRP
- **action**: ( )
- **reward**: $r_t = \alpha\, w_u R \;-\; \beta\, H \;-\; \gamma\, PP$
  - R: throughput, H: handover penalty, PP: ping-pong penalty
- clipped surrogate objective(PPO)

A1 xApp , xApp live E2 load/QoS .

5

5.1

MATLAB 5G Toolbox. O-CU 1 / O-DU 3 / O-RU 27, UE 100 , 7 km² . 2.1 GHz, 20 MHz, 46 dBm, UMa path loss, 8 dB shadowing. Fronthaul 10 Gbps (0.1–0.5 ms). Control interval 200 (20) 30 , 95% CI.

: Traditional A3 / O-RAN ML-Assisted(LSTM transfer learning) / Load Balance O-RAN.

5.2 KPI

	MultiModal-AHC	Traditional A3	ML-Assisted	Load Balance
Throughput (Mbps)	77.10 ± 5.72	74.11	78.48	75.23
HO rate	0.0362	0.0962	0.0209	0.4013
Ping-pong (%)	**0.00**	0.00	12.64	11.29

5.3

(MC)	k-NN	99.84%
(TP)	Random Forest	RMSE 14.40 m, 1.00
RSRP (P-RSRP)	RF ensemble	RMSE 7.35 dB
Cell Ranker (CR)	PPO Actor–Critic	reward +23.4, HO failure −18%, 150–180 ep

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stability–agility . A3 TTT ping-pong 0 , · RSRP ping-pong 0 . , throughput baseline CI .

6

. .

. MC 99.84%, 1.00, RSRP −66.76 vs −66.75. MATLAB kinematic trace , . testbed future work .

PPO . Abstract · Sec IV PPO rApp(non-RT) xApp , Sec III-C Actor–Critic "real-time decision engine" . non-RT real-time .

. "HO rate = HO failure rate" , MultiModal-AHC HO rate ML-Assisted proactivity . "18% failure " KPI PPO reward . throughput **ping-pong + HO rate** .

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7 & thesis

thesis .

7.1

- **building block** : P-RSRP( ) TP( ) thesis channel quality prediction · traffic prediction .
- : $(v, a, j, \dot{\psi})$ mobility mode thesis (/ AMR /) . ped / car / drone .
- **predict-then-decide** : proactive thesis blueprint **Stage 4(chance-constrained PRB)** .

7.2 (foil)

non-RT rApp + near-RT xApp A1/E2/O1 **RIC** . sub-second .

thesis .

Handover / mobility	sub-second	RIC (rApp/xApp, A1/E2)	
Importance-aware MAC scheduling	sub-TTI (µs)	collocation	E2

importance-aware MAC scheduling sub-TTI E2 xApp , GPU-collocated RAN+AI(pyAerial + GH200 + cuMAC) . "predictive mobility RIC " , .

7.3 ( )

Predictive mobility RIC (: MultiModal-AHC, rApp→A1→xApp).
sub-TTI importance-aware MAC scheduling — E2 near-RT (10 ms)
TTI (µs) . , GPU-collocated RAN+AI sub-TTI
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7.4 AI-RAN

predict-then-allocate sub-TTI · PRB · collocation thesis novelty (Stage 3 · 4)
. GH200 + pyAerial(CU · MAC/L1) cuMAC , ROS2 .

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8 take-away

1. sub-second RIC .
2. · + + RL predict-then-decide .
3. thesis sub-TTI · PRB , RIC .